Abalone Measurements and Shell Rings

Exploration, Statistical Inference and Regression

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Theory N4 · VibeCoding_Expert
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Theory endterm · group-confirmed submission details

Report video

YouTube report

https://youtu.be/D-H3yQ4IDro?si=VZ5VouVBJcCDhlBc

Planned duration: 12 minutes
Required duration: 7–15 minutes
Video deadline: end of 21 October 2026

Published report video · link confirmed by the group

Dataset and research questions

Property	Documented source
Observations / columns	4,177 / 9
Predictors / response	8 original inputs / Rings
Groups	M: 1,528 · F: 1,307 · I: 1,342
Stored measurements	Scaled dimensions and weights
Licence	CC BY 4.0

One dataset connects distributions, group inference, correlation and prediction.

Quality checks and holdout design

01 Preserve raw data; audit two zero heights

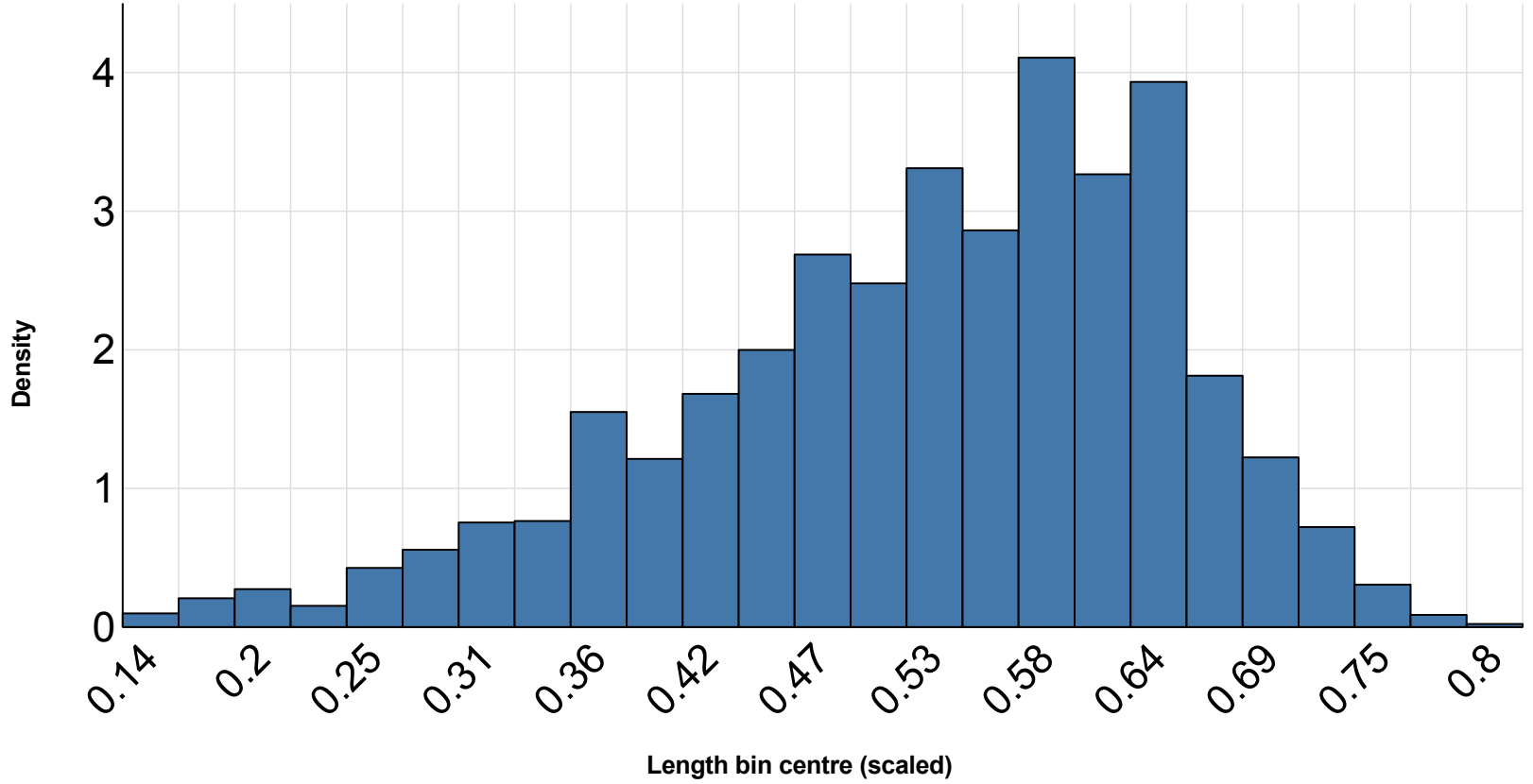
02 Training: 3,341 · holdout: 836 · seed 42

03 Fit imputation and scaling within each training fold

04 Flag unusual values; retain them in the primary model

The test set does not guide preprocessing, feature selection or deletion.

Distribution of recorded length

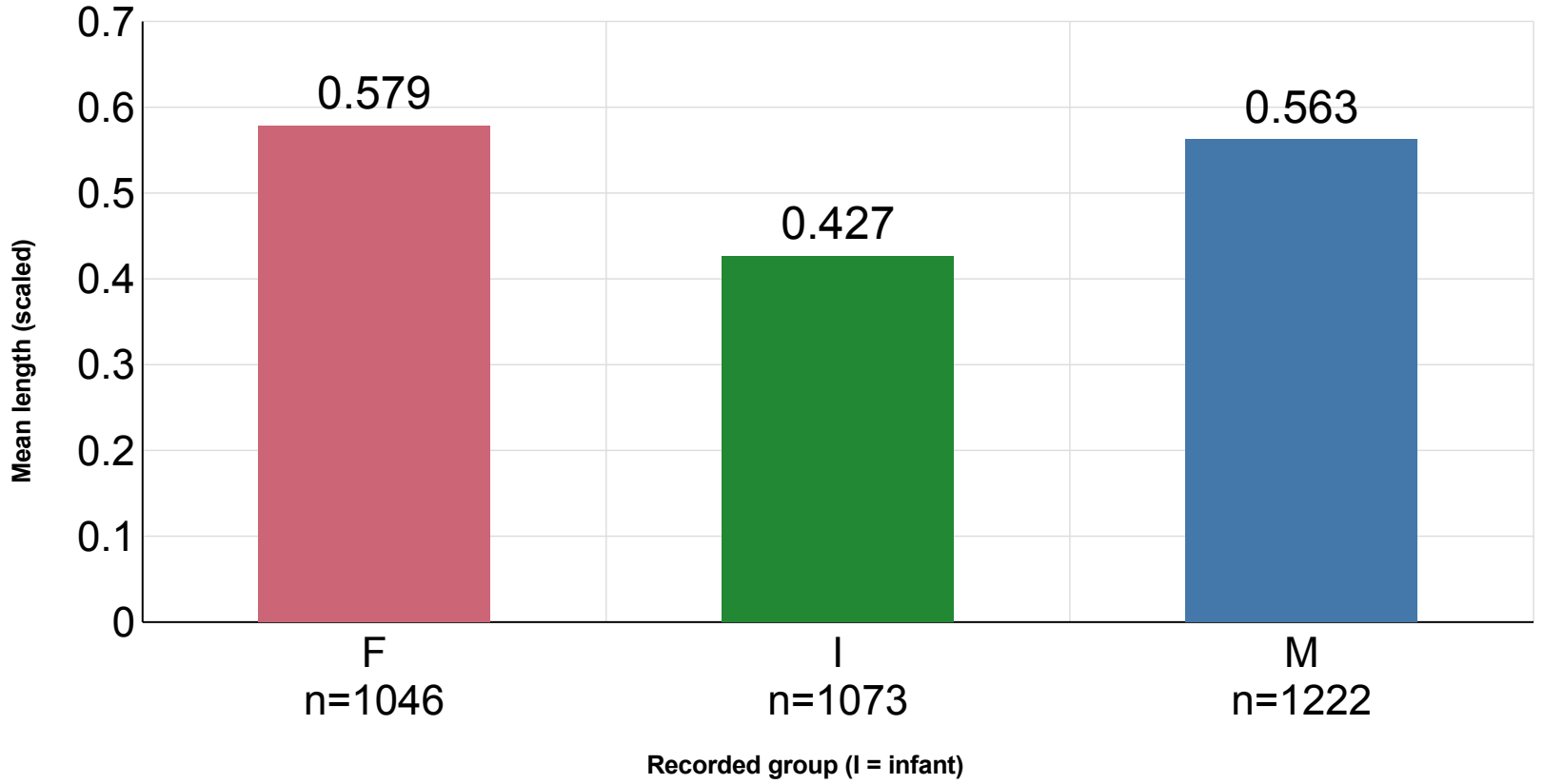


Training n = 3,341

Mean: 0.524
Median: 0.545
SD: 0.120

A smooth physical measurement need not follow a Normal distribution.

Length differs across recorded groups

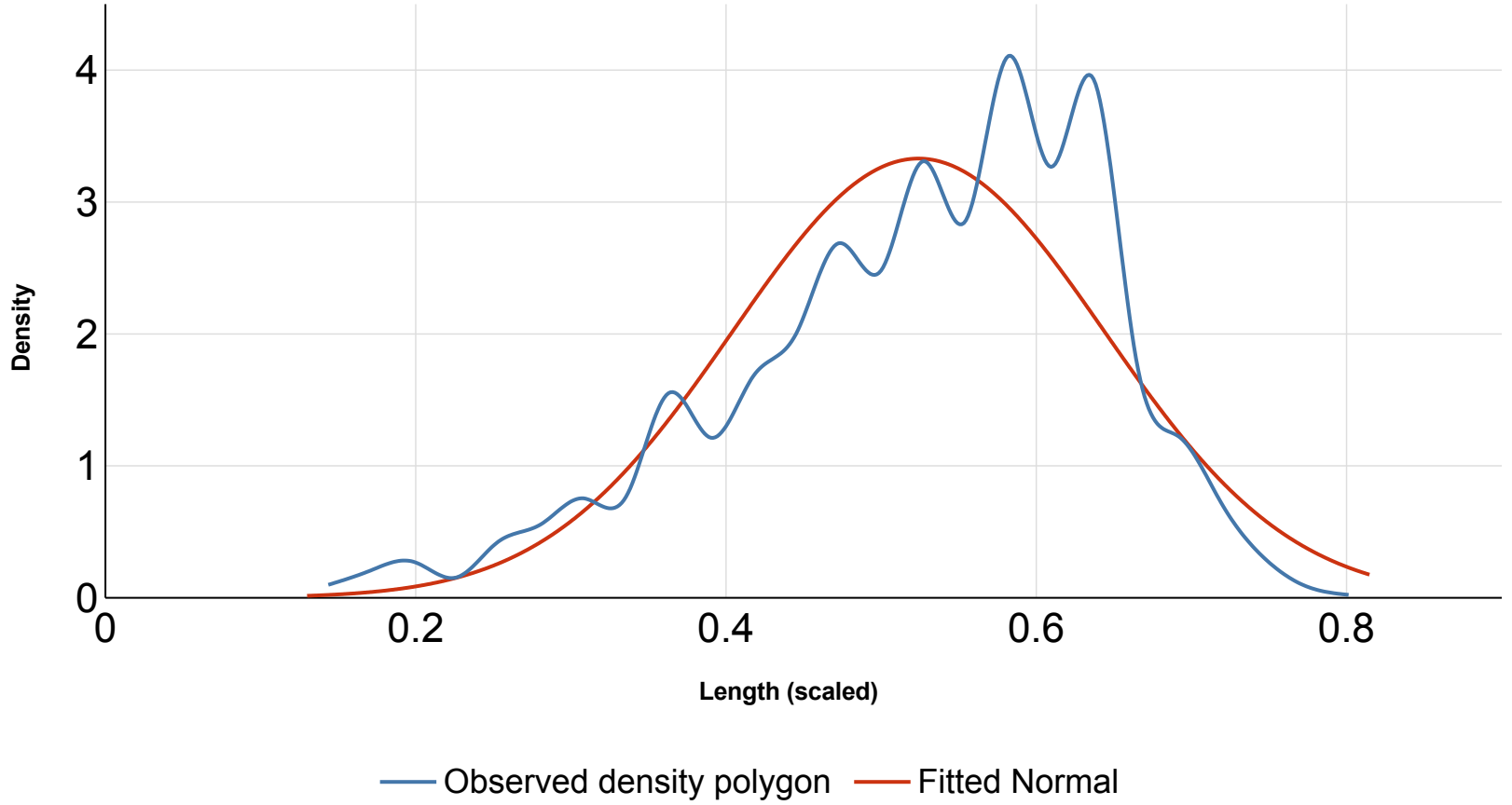


I means infant

Group comparisons mix biological sex and developmental status.

Observed group differences do not establish a causal mechanism.

Length against a Normal candidate

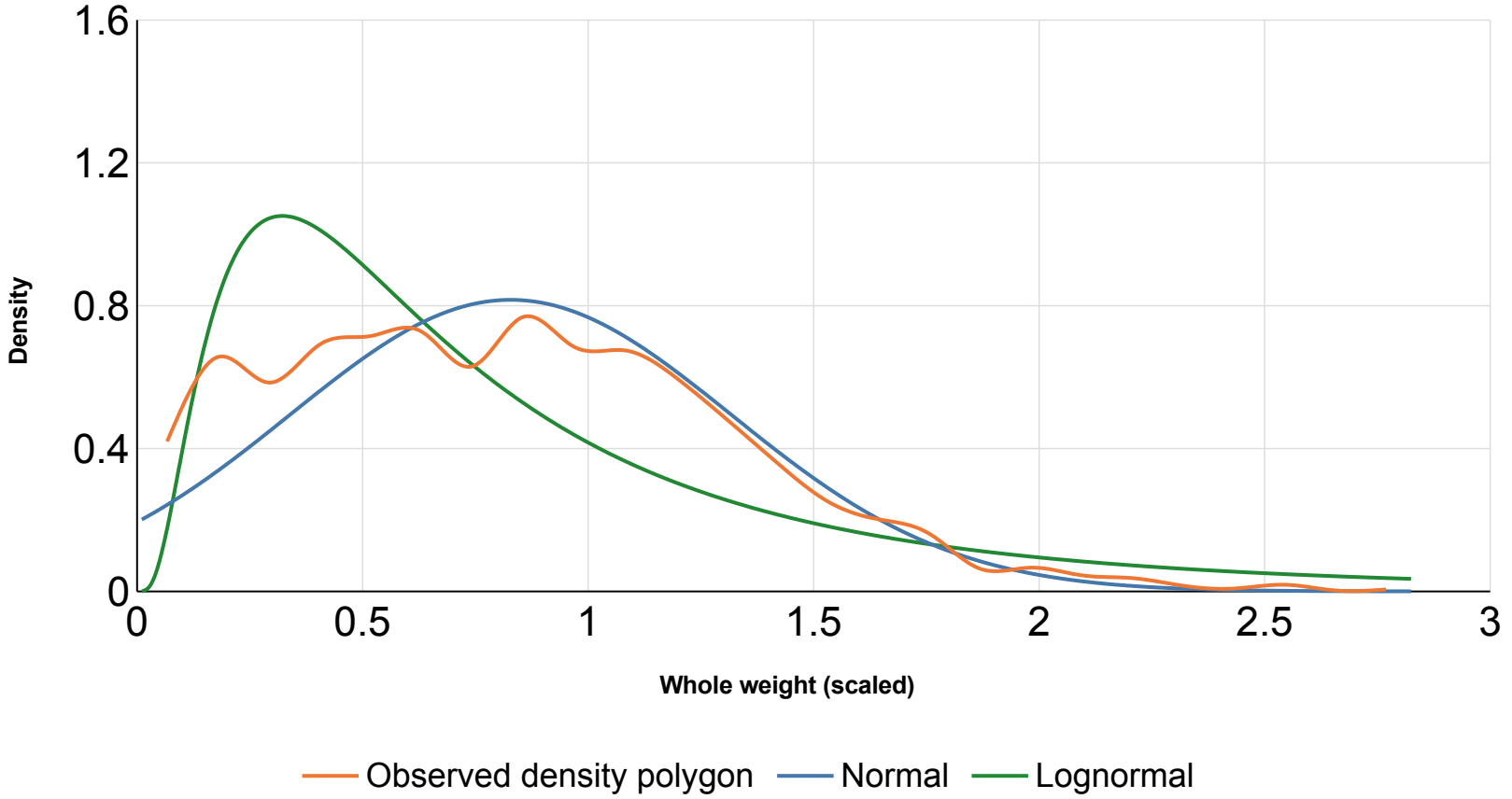


Normal rejected

AD = 29.40
Holm p = 0.0015
1,999 simulations

The Monte Carlo p-value reaches its resolution floor of 0.0005 before Holm.

Weight and candidate distributions



Both rejected

Normal AD: 16.47
Lognormal AD: 86.51

Holm p = 0.0015

Rejection is a valid finding; no extra families are searched to obtain a preferred p-value.

Three prespecified hypothesis tests

Test	Null hypothesis
Welch t	Mean Rings(M) = mean Rings(F)
Chi-square	Group and ring band are independent
Welch ANOVA	Mean Length(F) = mean Length(I) = mean Length(M)

$\alpha = 0.05$ · Holm correction across the three primary questions

Methods still assume appropriate sampling independence; missing batch IDs limit inference.

Results and effect magnitude

Test	Statistic	Holm p
Welch t	-2.765	0.0057
Chi-square	925.83	8.45e-199
Welch ANOVA	720.44	6.09e-241

M − F: -0.355 rings
95% CI [-0.606, -0.103]

Cramér's V = 0.372
Minimum expected count: 334.7

Statistical significance and practical magnitude answer different questions.

Pearson and Spearman associations

Spearman ρ

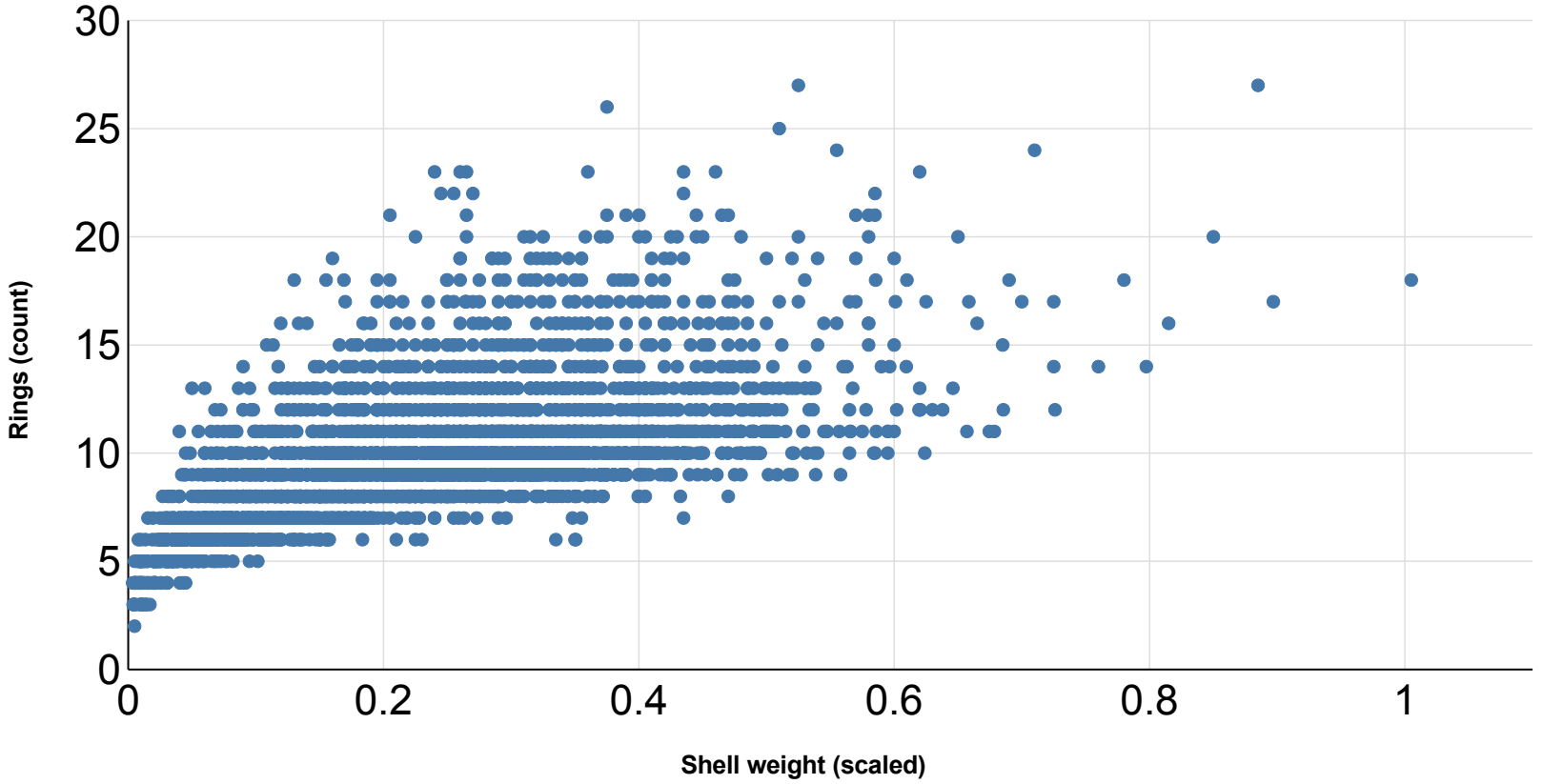


Shell weight ↔ Rings

Pearson r = 0.628
Spearman ρ = 0.693
n = 3,341

High correlation among measurements can destabilize conditional regression slopes.

Shell weight and Rings



$\rho = 0.693$

Marginal association

Integer rings create overlapping points.

This relationship is descriptive; it is not an independent causal effect.

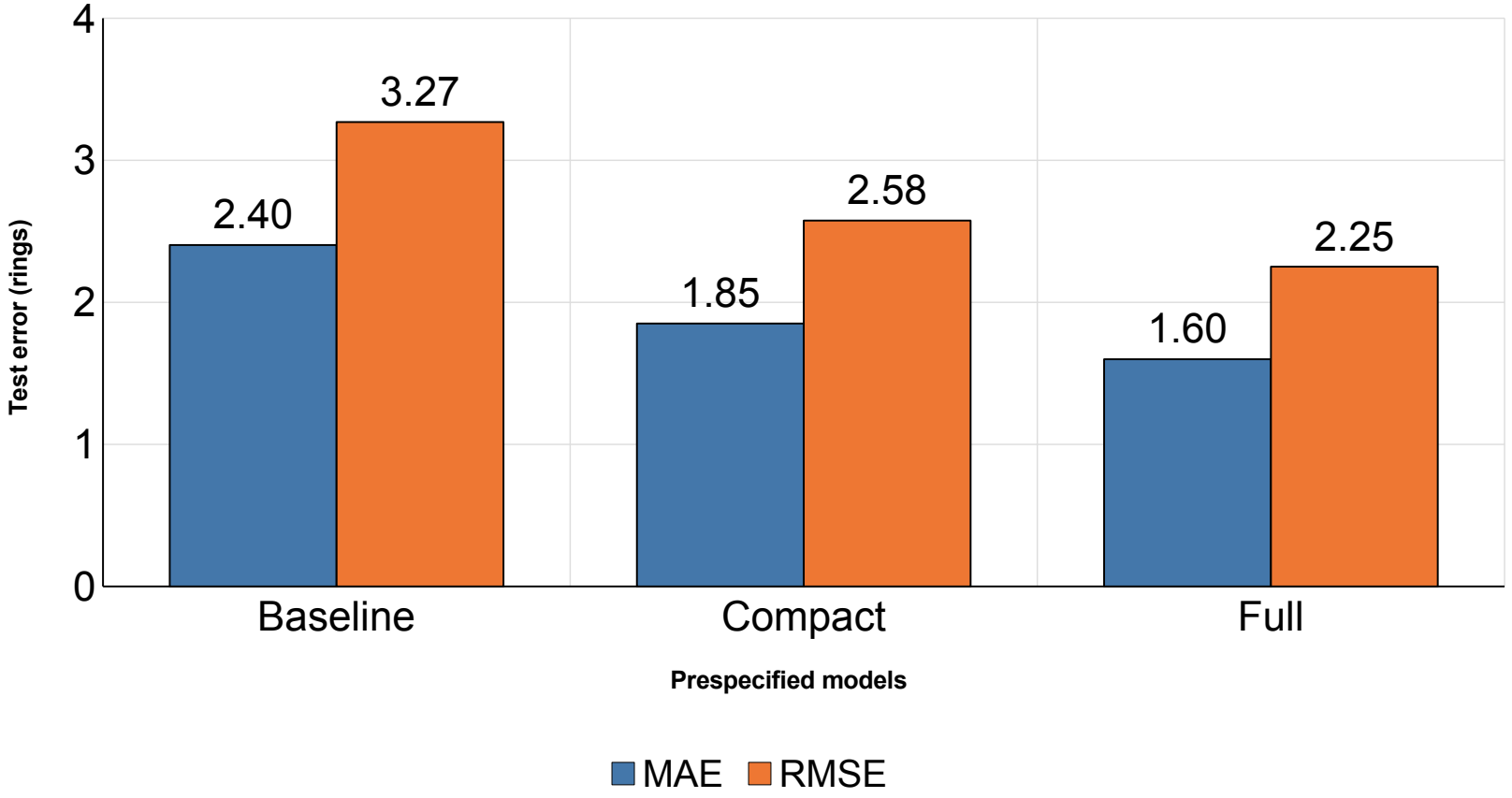
Multiple linear regression design

$$\text{Rings} = \beta_0 + \Sigma\, \beta_j x_j + \varepsilon$$

Model	Inputs
Baseline	Training mean only
Compact	Group + dimensions + whole weight
Full	All eight original inputs

Fivefold CV fits a new imputer, scaler and regression within every training fold.

Holdout performance and model limits



Full R^2 = 0.526

RMSE: 2.25 rings
31.1% below baseline

One out-of-range prediction

High VIF, heteroskedasticity and influence limit coefficient interpretation.

Reproducible code demonstration

01 Verify source checksum and saved train/test indexes

02 Inspect hypotheses, confidence interval and Holm correction

03 Show preprocessing inside CV and the test prediction CSV

04 Trace report and slide numbers back to the exported summary

The executed notebook remains available as the offline demonstration fallback.

Conclusions and limitations

Distribution candidates rejected

Full-model test MAE: 1.60 rings

Historical sampling, collinearity and residual patterns limit generalization.

Observed association and predictive improvement do not establish biological causation.

Appendix: definitions and inference

01 Welch t: mean difference / estimated standard error

02 Chi-square: Σ (observed − expected)^2 / expected

03 Classical ANOVA: MS between / MS within

04 Holm: control the three prespecified primary questions

Appendix: regression diagnostics

Check	Observed evidence
Design rank	10 / 10
Breusch–Pagan p	1.44e-78
Cook > 4/n	212 training records
Full coefficient covariance	HC3; numeric slopes per training SD
Sensitivity design	Filter inner train; keep validation unchanged

The primary model retains influential observations; sensitivity is reported separately.

References

Nash et al. (1994). Abalone. UCI Machine Learning Repository.
https://doi.org/10.24432/C55C7W

SciPy 1.15.3 — statistical tests and goodness of fit
https://docs.scipy.org/doc/scipy-1.15.3/

Statsmodels — Welch ANOVA, HC3 and influence
https://www.statsmodels.org/

Scikit-learn 1.6 — preprocessing and validation
https://scikit-learn.org/1.6/

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